

Firewall Log Analysis with Splunk

This case study focused on analyzing a simulated firewall log file using Splunk to identify suspicious network behavior, understand traffic flow, and assess potential security threats. The dataset mimicked real-world firewall activity and included various types of events such as allowed and blocked connections, intrusion detection system (IDS) alerts, access attempts to high-risk ports, and repeated requests from the same IP addresses.

Data Ingestion and Field Extraction

The firewall log file (`firewall_log_example.txt`) was ingested into Splunk using the "Upload File" feature. During ingestion, efforts were made to ensure proper parsing of log entries and the automatic or manual extraction of key fields such as:

- `timestamp`: When the event occurred
- `src_ip`: Source IP address initiating the connection
- `dest_ip`: Destination IP address targeted by the request
- `dest_port`: Port number being accessed
- `protocol`: Network protocol used (e.g., TCP, UDP)
- `action`: Whether the connection was allowed or blocked
- `reason`: The reason for blocking or IDS trigger (if any)

These fields provided the foundation for detailed search and analysis in Splunk.

Initial Log Exploration

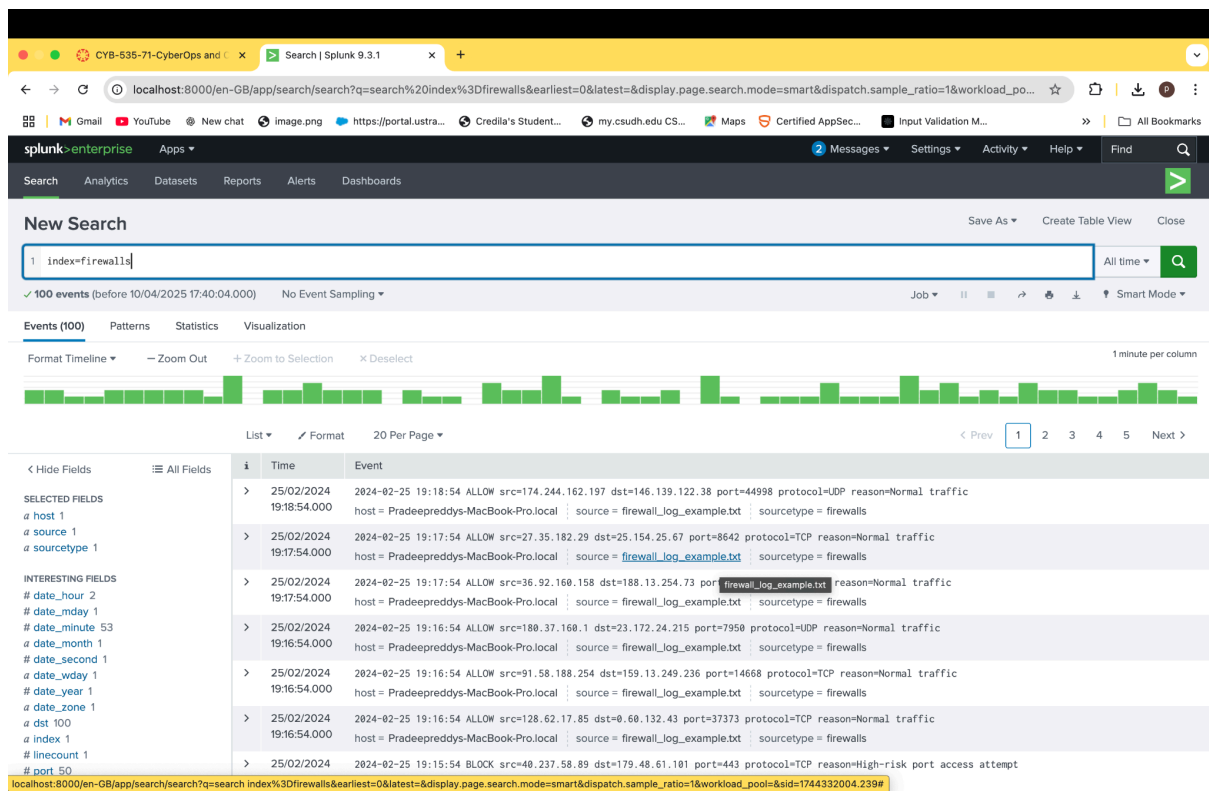
An initial examination of the log entries revealed a diverse set of network events. The distribution between **allowed** and **blocked** traffic was analyzed to get a high-level understanding of normal vs. potentially malicious activity. The most frequently used **protocols** and **destination ports** were identified, along with a list of top source IP addresses generating network traffic.

Security Threat Identification

Several potential security threats were uncovered through deeper analysis:

- **Repeated access attempts** from specific IPs, suggesting brute-force or scanning behavior.
- **High volumes of blocked traffic targeting sensitive ports** like 22 (SSH), 23 (Telnet), and 3389 (Remote Desktop), which are often exploited in attacks.
- **Frequent IDS alerts** were logged, highlighting attempted connections that matched known attack patterns or policy violations.

A time-based breakdown of events helped identify whether these threats occurred in bursts, during off-hours, or followed consistent patterns — all of which are valuable indicators of automated or targeted attacks.



CYB-535-71-CyberOps and xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewalls%20%7C%20top%20protocol&earliest=0&latest=&display.page.search.mode=smart&dispatch....

GmailYouTubeNew chatimage.pnghttps://portal.ustra...Credlia's Student...my.csudh.edu CS...MapsCertified AppSec...Input Validation M...All Bookmarks

splunk>enterpriseAppsMessagesSettingsActivityHelpFindSearchAnalyticsDatasetsReportsAlertsDashboards

New SearchSave AsCreate Table ViewClose

1 index=firewalls | top protocolAll time

✓ 100 events (before 10/04/2025 17:41:39.000)No Event SamplingJobSmart Mode

EventsPatternsStatistics (2)Visualization

20 Per PageFormatPreview

protocol	count	percent
UDP	56	56.000000
TCP	44	44.000000

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localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewalls%20%7C%20top%20dest_port&earliest=0&latest=&display.page.search.mode=smart&dispatc...

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splunk>enterpriseAppsMessagesSettingsActivityHelpFindSearchAnalyticsDatasetsReportsAlertsDashboards

New SearchSave AsCreate Table ViewClose

1 index=firewalls | top dest_portAll time

✓ 100 events (before 10/04/2025 17:42:18.000)No Event SamplingJobSmart Mode

EventsPatternsStatistics (0)Visualization

20 Per PageFormatPreview

No results found.

CYB-535-71-CyberOps and xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewalls%20%7C%20stats%20count%20by%20src_ip%20%7C%20sort%20-count&earliest=0&latest=...

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splunk>enterpriseAppsMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

Save AsCreate Table ViewClose

1 index=firewalls | stats count by src_ip | sort -count

All time

✓ 100 events (before 10/04/2025 17:42:46.000)No Event SamplingJob

EventsPatternsStatistics (0)Visualization

20 Per PageFormatPreview

No results found.

CYB-535-71-CyberOps and xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewalls%20%7C%20top%20dest_port&earliest=0&latest=&display.page.search.mode=smart&dispatc...

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splunk>enterpriseAppsMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

Save AsCreate Table ViewClose

1 index=firewalls | top dest_port

All time

✓ 100 events (before 10/04/2025 17:42:18.000)No Event SamplingJob

EventsPatternsStatistics (0)Visualization

20 Per PageFormatPreview

No results found.

CYB-535-71-CyberOps and xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewalls%20%7C%20timechart%20span%3D1h%20count%20by%20dest_port&earliest=0&latest=&di...

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SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

Save AsCreate Table ViewClose

1 index=firewalls | timechart span=1h count by dest_portAll time

✓ 100 events (before 10/04/2025 17:43:14.000)No Event SamplingJobVisualization

EventsPatternsStatistics (2)Visualization

20 Per PageFormatPreview

_time	NULL
2024-02-25 18:00	64
2024-02-25 19:00	36

CYB-535-71-CyberOps and xSearch | Splunk 9.3.1x+

localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewalls%20alert_type%3D%20OR%20reason%3DIDS%20Alert*&earliest=0&latest=&display.page.s...

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splunk>enterpriseAppsMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

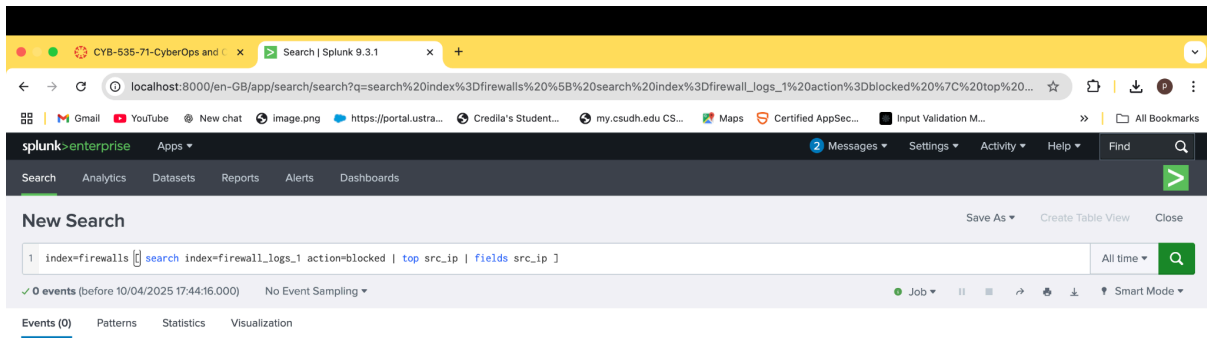
Save AsCreate Table ViewClose

1 index=firewalls alert_type=* OR reason="IDS Alert"All time

✓ 0 events (before 10/04/2025 17:43:41.000)No Event SamplingJobVisualization

Events (0)PatternsStatisticsVisualization

No results found.



Visualization and Dashboarding

To better illustrate the findings, several visual panels were created and added to a custom Splunk dashboard titled **"Firewall log"**. The dashboard included:

- A pie chart showing the ratio of allowed to blocked traffic
- A timechart visualizing the frequency of blocked attempts across time intervals
- A bar chart showing the top 10 most targeted ports
- A table summarizing IDS alerts and the source IPs associated with them

These visualizations made it easier to interpret trends, spot anomalies, and report findings clearly.

CYB-535-71-CyberOps and xfirewall log | Splunk 9.3.1 x

localhost:8000/en-GB/app/search/firewall_log?form.global_time.earliest=-24h%40h&form.global_time.latest=now

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splunk>enterpriseAppsMessagesSettingsActivityHelpFindSearchAnalyticsDatasetsReportsAlertsDashboards

firewall logActionsEdit

Global Time RangeLast 24 hours

2024-02-25T20:00:00.000-08:000

2024-02-25T21:00:00.000-08:000

2024-02-25T22:00:00.000-08:000

2024-02-25T23:00:00.000-08:000

< Prev12345... Next>

protocol	count	percent
UDP	56	56.000000
TCP	44	44.000000

< Prev12345... Next>

i	Time	Event
>	2/25/2024	2024-02-25 19:18:54 ALLOW src=174.244.162.197 dst=146.139.122.38 port=44998 protocol=UDP reason=Normal traffic